



# Unsupervised Machine Learning Model for Characterizing Phishing Attacks

Juan Esteban Diaz Castro

Universidad de los Andes  
Departamento de Ingeniería de Sistemas y Computación  
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**Juan Esteban Diaz Castro**

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Directores:  
Ph.D. Carlos Andrés Lozano Garzón

Universidad de los Andes  
Departamento de Ingeniería de Sistemas y Computación  
Ingeniería de Sistemas y Computación  
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2025



## Dedictory

We can only see a short distance ahead, but we can see plenty there that needs to be done.

Alan Turing



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I want to thank my father, mother, sister, and my dog Loky, who have been a constant support throughout the whole of my career and the realization of this undergraduate thesis. Also, I want to thank the Professor Carlos Andrés Lozano Garzón for his help and advice for developing my thesis and all the content found inside of it.





# Contents

|          |                                          |           |
|----------|------------------------------------------|-----------|
| <b>1</b> | <b>List of Figures</b>                   | <b>8</b>  |
| <b>2</b> | <b>List of Tables</b>                    | <b>9</b>  |
| <b>3</b> | <b>List of Symbols and Abbreviations</b> | <b>10</b> |
| <b>4</b> | <b>Abstract</b>                          | <b>11</b> |
| <b>5</b> | <b>Resumen</b>                           | <b>12</b> |
| <b>6</b> | <b>Introduction</b>                      | <b>13</b> |
| 6.1      | Problem Context . . . . .                | 13        |
| 6.2      | Problem Justification . . . . .          | 15        |
| 6.3      | Objectives . . . . .                     | 17        |
| 6.3.1    | General Objectives . . . . .             | 17        |
| 6.3.2    | Specific Objectives . . . . .            | 17        |
| 6.4      | Limitations . . . . .                    | 17        |
| <b>7</b> | <b>Theoretical Framework</b>             | <b>19</b> |
| 7.1      | Machine Learning . . . . .               | 19        |
| 7.1.1    | Deep Learning . . . . .                  | 19        |
| 7.1.2    | How Does it Work? . . . . .              | 19        |
| 7.1.3    | Supervised Learning . . . . .            | 20        |
| 7.1.4    | Reinforced Learning . . . . .            | 22        |
| 7.1.5    | Unsupervised Learning . . . . .          | 23        |
| 7.2      | Unsupervised Learning . . . . .          | 25        |
| 7.2.1    | Clustering . . . . .                     | 26        |
| 7.2.2    | Exclusive Clustering . . . . .           | 26        |
| 7.2.3    | Overlapping Clustering . . . . .         | 27        |
| 7.2.4    | Hierarchical Clustering . . . . .        | 28        |
| 7.2.5    | Probabilistic Clustering . . . . .       | 29        |
| 7.2.6    | Association . . . . .                    | 29        |
| 7.2.7    | Dimensionality Reduction . . . . .       | 30        |
| 7.2.8    | Scikit-Learn . . . . .                   | 30        |
| 7.2.9    | Algorithms . . . . .                     | 30        |
| 7.3      | Cybercrimes . . . . .                    | 31        |
| 7.3.1    | Phishing . . . . .                       | 32        |
| 7.4      | Social Engineering . . . . .             | 34        |
| 7.4.1    | Social Engineering on Phishing . . . . . | 35        |
| <b>8</b> | <b>Metodology</b>                        | <b>36</b> |
| 8.1      | Approach . . . . .                       | 36        |
| 8.2      | Research Design . . . . .                | 36        |
| 8.3      | Data Collection . . . . .                | 36        |

|           |                                                  |           |
|-----------|--------------------------------------------------|-----------|
| <b>9</b>  | <b>Development &amp; Proposal</b>                | <b>37</b> |
| 9.1       | Preliminary Model . . . . .                      | 37        |
| 9.1.1     | Data Structure . . . . .                         | 37        |
| 9.1.2     | Data Preparation & Preliminary Results . . . . . | 37        |
| 9.1.3     | First Conclusions . . . . .                      | 40        |
| 9.2       | Final Model . . . . .                            | 40        |
| 9.2.1     | Data Structure . . . . .                         | 41        |
| 9.2.2     | Data Understanding . . . . .                     | 41        |
| 9.2.3     | Data Processing . . . . .                        | 42        |
| 9.2.4     | Modeling . . . . .                               | 42        |
| <b>10</b> | <b>Results</b>                                   | <b>44</b> |
| 10.1      | Model . . . . .                                  | 44        |
| 10.1.1    | Clusters . . . . .                               | 44        |
| 10.1.2    | K-Means Metrics . . . . .                        | 45        |
| 10.1.3    | Cluster Characterization . . . . .               | 48        |
| 10.2      | User Interface . . . . .                         | 49        |
| <b>11</b> | <b>Discussion</b>                                | <b>51</b> |
| 11.1      | Phishing Attacks Characteristics . . . . .       | 51        |
| 11.2      | Machine Learning Models . . . . .                | 51        |
| 11.3      | K-Means Machine Learning Model . . . . .         | 51        |
| 11.4      | Model Metrics . . . . .                          | 52        |
| 11.5      | Cluster Analysis . . . . .                       | 52        |
| <b>12</b> | <b>Conclusions</b>                               | <b>53</b> |
| <b>13</b> | <b>Appendices</b>                                | <b>55</b> |
| <b>14</b> | <b>Bibliography</b>                              | <b>56</b> |

# 1 List of Figures

|    |                                                              |    |
|----|--------------------------------------------------------------|----|
| 1  | Linear and Polynomial Regression Algorithms [16] . . . . .   | 21 |
| 2  | Support Vector Machine Algorithm [17] . . . . .              | 21 |
| 3  | Decision Tree Algorithm [18] . . . . .                       | 22 |
| 4  | Reinforced Learning [19] . . . . .                           | 23 |
| 5  | K-Means Algorithm [20] . . . . .                             | 24 |
| 6  | Cluster Dendrogram [21] . . . . .                            | 25 |
| 7  | Hierarchical Clustering Algorithm [21] . . . . .             | 25 |
| 8  | Exclusive Clustering [26] . . . . .                          | 27 |
| 9  | Exclusive vs Overlapping Clustering [27] . . . . .           | 28 |
| 10 | Agglomerative Clustering - Ward [28] . . . . .               | 28 |
| 11 | Probabilistic Clustering - GMM [29] . . . . .                | 29 |
| 12 | Most Phished Brands on Q1 2024 [13] . . . . .                | 32 |
| 13 | Example of a Phishing Email [33] . . . . .                   | 35 |
| 14 | Results Obtained with the Bag of Words Strategy . . . . .    | 38 |
| 15 | Results Obtained with the Text Embeddings Strategy . . . . . | 39 |
| 16 | Emails Word Cloud . . . . .                                  | 42 |
| 17 | Clusters Generated With K-Means . . . . .                    | 44 |
| 18 | Clusters Members . . . . .                                   | 45 |
| 19 | Elbow Method . . . . .                                       | 46 |
| 20 | Silhouette Coefficients for different $k$ values . . . . .   | 47 |
| 21 | Silhouettes for the $k = 3$ groups . . . . .                 | 48 |
| 22 | User Interface Elaborated for Presenting the Model . . . . . | 50 |
| 23 | User Interface Limitations for Empty Data . . . . .          | 50 |

## 2 List of Tables

|   |                                                       |    |
|---|-------------------------------------------------------|----|
| 1 | Clustering Algorithms Comparison . . . . .            | 31 |
| 2 | Clustering Preliminary Notebooks Comparison . . . . . | 40 |
| 3 | Clustering Data Partition . . . . .                   | 45 |
| 4 | $k$ values metric comparison . . . . .                | 47 |

### 3 List of Symbols and Abbreviations

**AI** Artificial Intelligence. 13, 14, 19, 20, 34

**ASCII** American Standard Code for Information Interchange. 37, 42

**BEC** Business Email Compromise. 34

**BoW** Bag of Words. 20, 38

**DBSCAN** Density-Based Spatial Clustering of Applications with Noise. 26, 40, 51, 55

**DDoS** Distributed Denial of Service. 15, 16, 31

**DL** Deep Learning. 19

**EAC** Email Account Compromise. 34

**GMM** Gaussian Mixture Models. 8, 29, 40, 51

**GPU** Graphics Processing Units. 19

**HCA** Hierarchical Cluster Analysis. 28

**HDBSCAN** Hierarchical Density-Based Spatial Clustering of Applications with Noise. 26, 31, 40, 51, 55

**LLM** Large Language Models. 19

**ML** Machine Learning. 11, 12, 14, 15, 19, 20, 30, 40

**OS** Operating System. 16

**SVM** Support Vector Machine. 21, 22

**VoIP** Voice over IP. 34

## **4 Abstract**

The present undergraduate thesis work is focused on the development of an unsupervised Machine Learning (ML) model for characterizing Phishing attacks. For achieving this, the main aim was to implement a K-Means algorithm model to train with external email data for later on generating the clusters according to their email content with the help of Text Embeddings. Additionally, with the help of this model, the objective was to study the main characteristics associated with the clusters generated from these emails. In the end, the implemented model helps to understand some of the characteristics found inside Phishing emails, like length, URL presence and also, focuses on the contextual analysis by studying the words found inside each cluster.

## **Key Words**

Characteristics, Clustering, Emails, Embeddings, K-Means, Machine Learning, Model, Phishing, Train, Unsupervised, URL & Words

## 5 Resumen

El presente trabajo de tesis de pregrado se centra en el desarrollo de un modelo de Machine Learning (ML) para poder caracterizar ataques de Phishing. Para poder lograr esto, el objetivo principal fue implementar un modelo con el algoritmo de K-Means entrenado con un conjunto de datos sobre emails por medio de la tecnica de Embeddings. Adicionalmente, con la ayuda de este modelo se enfocó en poder estudiar las características principales asociadas con todos los grupos generados a partir de los datos estudiados. Al final, el modelo implementado ayuda a entender algunas de las características que los emails de Phishing contienen, como su longitud, presencia de URL y también, se enfoca en el analysis contextual estudiando las palabras encontradas dentro de cada cluster.

### Palabras Claves

Agrupación, Aprendizaje de Máquina, Características, Emails, Embeddings, Entrenamiento, K-Means, Modelo, No Supervisado, Palabras, Phishing & URL

## 6 Introduction

The following thesis work is focused on the implementation of a Machine Learning model in the context of Cybersecurity, the reason for working in these two areas (Cybersecurity and Machine Learning), is due to the fact that Cybercrimes have become so common in the actuality and using Machine Learning enables the possibility to understand some deep characteristics of the most frequent digital attacks performed by the criminals. Along the document there is going to be a theoretical analysis, an experimental setup with the work proposal and finally some results analysis and conclusions regarding the most important aspects found on the elaborated prototype.

### 6.1 Problem Context

Now on days, technology has advanced pretty fast and has accessed new boundaries society could only dream of years ago, all these discoveries and achievements have helped massively the development of humankind, since the discovery of Internet to the point where Artificial Intelligence (AI) is commonly used at the time, for helping in simple tasks, each of these achievements have left their mark in human history. However, the full development of technology has its disadvantages, being the most common one the abuse of technology at peoples disposal for unethical purposes and harming others. In the digital world, this is known as Cybercrimes and it is one of the most common problems this world has ever faced since many years ago, and at the present time there is still no viable solution for it, thus it continues to be important to try to learn how to face these harms the technology brings society. The real matter is that it is so easy to abuse and use technology for incorrect purposes, taking into account that Cybercrimes can be committed by simply behaving properly on the Internet or by stealing people's private information. Inside this huge crime ocean, the most common ones are the attacks in which malware, ransomware data kidnap and data lurk are involved.

Easily, it could be said that the optimal solution for the current Cybercrimes is the establishment of navigation and Internet consumption policies, that could stop these unethical acts; however, the main problem is the fact that the Internet is a wide digital space that has no central operational site like a server. Thus, this means that focusing on something that has no central body would be almost impossible to control and regulate, implying that policies are not a worthy solution as there is no guarantee that they would work out. Nevertheless, there are some considerations that can be used for protection towards the information systems, services, private data and users when faced with one of the different threats that exist on the Internet.

In other words, it is important to talk about this topic since Cybercrimes have drastic consequences among all digital users, not only businesses. All of this according to the World Economic Forum and several Cybersecurity orga-



nizations which state that only in 2024, 9 out of 10 businesses were victims of cybernetic threats which implied Million-dollar losses, additionally, according to them and an estimation of the year 2025, the monetary losses could go up to Billion-dollar losses. Furthermore, in 2024 the number of weekly attacks registered was 1,876 per organization, this represented a growth of 75% from the cases recorded in 2023. Also, it is important to consider that many of the attacks inside this 75% growth are now AI generated, due to the easy personalization and efficiency options it gives to the users, thus making the attacks more difficult to detect as they are built different from the common ones, that have been used for many years, and try to lurk the user by making them believe the altered attack form is 100% true. [8]

On the other hand, at the current time (2025) the expenses caused by Cybercrimes add up to a total of 10.5 billion dollars, making this amount bigger than the GDP of many countries. These expenses are arising because, inside the huge number of affected businesses, only 20% of these are prepared to manage and solve the different threats that affect them. However, the amount of money invested in Cybercrimes has not been reduced with the passage of years, a particular case that demanded a huge number of monetary resources was the 2017 ransomware WannaCry attack. By its part, WannaCry attack infested almost 230,000 devices on 150 different countries and meant a total of 4 thousand million dollars in damages, making this one of the worst scenarios that Cybercrimes has caused. [3][8]

Because of this, there are different tools used for discovering threats such as firewalls and anti-viruses, even so there is the possibility to use Machine Learning, which is a branch of AI and enables the users to make brief synthesis of data. Moreover, as stated by Cisco, using Machine Learning enhances the possibility to have different analysis of a network data searching for any type of anomalies inside the data, in many cases these anomalies mean possible malware or suspect threats. Besides, Machine Learning is used to make a quick study of the neighbors in a network so the user can avoid contacting and/or connecting with suspicious endpoints or sites that can imply an attack, as they make the user believe they are safe contacting with them. Also, in many cases, Machine Learning is used to do a file review, as it checks the origin and characteristics of many multimedia files that navigate through the Internet, checking for threats such as malware inside of these. [2]

Meanwhile, many of the current existing models for traffic detection are trained through supervised learning and need data previously categorized in order to work with future data, there also exist the possibility to have models that are created using the unsupervised learning techniques, and because of that there are few unsupervised ML models elaborated for detecting anomalies. Alternatively, unsupervised Machine Learning models are rarely used because they work with data with no labels; however, it is important to notice that these models work for finding out patterns and characteristics of datasets, in other

words, they work as a form of recognition of new threat patterns and thus they group data with similar characteristics into groups which enables them to have a base for possible harmful data, all of this enables the opportunity to draw patterns and associate them to a particular element. Lastly, the reinforced models are more complex as they are subject to trial and error, like human learning, they work out ideally for analyzing Distributed Denial of Service (DDoS), because they can learn from requests and conclude which ones are appropriate and safe, and which ones can imply on a server overload. [9]

Overall, the use of Machine Learning for the field of Cybersecurity is caused by the large number of traffic data a network has, because of this analyzing each data element by human thinking would limit the speed and ability to fully analyze data. For that reason using Machine Learning models helps to make a quick synthesis of the data regardless of its volume, and therefore, it makes it easier to analyze patterns and characteristics in order to conclude over a dataset and be able to find out any possible anomalies or strange patterns the data has. It is important to notice also that human interaction with data can be biased at some point, and it could imply that the analysis that is being done could not be completely impartial, thus using a ML model avoids the possible bias when analyzing data.

## 6.2 Problem Justification

For instance, the main victims of these digital threats are the different Internet users that exist, nonetheless, attackers tend to focus on the big companies and organizations like North Face, Washington Post, Cartier and more, as they have huge amounts of money that makes them subject of the hungry criminals. The main reason why it is important to take this subject into account is because there are different methods used for unethical activities on the Internet, and because of that attackers can always find a way to trick users and steal from them. For instance, the threats that report the higher numbers of cases are malware, ransomware, DDoS and social engineering, thus making them the most used for Cybercrimes on the Internet. [10]

Furthermore, and making a deeper emphasis on some of the cybernetic threats that exist, Malware by itself is the abbreviation for the term Malicious Software which is a code or program that has been designed with the purpose of harming systems or users that have direct contact with it. Additionally, Ransomware is a type of Malware which encrypts data and kidnaps it, making it unreadable for the owner of the data until a specific amount of money is paid, this is basically digital extortion, as criminals demand money for giving back the data. On the other hand, Phishing is the practice of sending fraudulent e-mails or messages that trick users into downloading Malware and sharing private and personal information, so that attackers can steal the identity of a person. Along with all the software threats, internal threats exist, and the most common ones are the DDoS in which there is a log of requests that are designed to saturate

and overload a server, so it is doomed to fail and collapse. Finally, there are also cases in which the cybercriminals own an authorized user and by using the credentials they are able to access the data and organization system, all of this is done without alerting the antivirus software or the firewalls, attacking and accessing the system through the interior, this is done after they obtained the credentials with the help of Social Engineering. Thus it is important to take into account that Social Engineering is one of the most common ways that cybercriminals fool Internet users into thinking that they are giving their personal information to the right people. [5]

In brief, the Internet is a very insecure place as there are plenty of forms and techniques that can be used for tricking, in order to obtain valuable information, all of this makes it a space where any user error could end up having several consequences. However, this could be better understood by looking at the statistics and numbers reported from some of the most common techniques used for treating users on the Internet. Thus, it is important to focus on the numbers the variety of methods for Cybercrimes generate, as it is essential to understand to its fullest why this is a serious problem.

According to the data recollected on AV-ATLAS for July of 2025, there are a total of approximately 977,000,000 types of malware that affect different devices with Windows Operating System (OS), in which Trojan viruses are the most common ones. Additionally, for the month of August there have been a total of 1,000 electronic mails declared as spam each day, for which 40% of this amount are declared as phishing. In addition, for the 14th of August there were registered around 57,000 cases of attacks to systems and infrastructure, this shows that for the date (2025), there is not enough security when online networking and due to the high number of threats there exist, the numbers of cases are too high. [1]

On the other hand, this is something that has been happening for a long time, as reported by IBM, between 2020 and 2021, there was an increase of 33% in attacks due to vulnerability exploitation. By 2020, the most common threat was the DDoS as in the first half of the year there were a total of 48.3 million attacks registered, which means an approximate of 26,000 attacks per day. Also, in the year 2021 ransomware attacks were the most common ones, representing a 21% of the total reported threats and in 4% of the cases, the victim companies paid for recovering the kidnapped data, adding to a total of 35 thousand million dollars. [3][6][7]

Above all, this shows that there is no unique way of attacking and harming all the Internet users, instead cybercriminals make use of different techniques to trick the users and make them their puppets as they end up following their instructions. Because of that, it is important to search for a solution which could study the different factors, actors and elements that exist on the Internet and as of that identify elements with a harmful nature, so that the users could avoid

possible digital threats. However, it is important to consider that elaborating a final solution for these problems represents a huge level of complexity, as due to the huge number of causes it is impossible to focus them all on a single solution, rather the elaboration of individual solutions is better as it can focus especially on an individual cause.

## **6.3 Objectives**

### **6.3.1 General Objectives**

The main objective of this thesis is to implement an unsupervised Machine Learning model for detecting possible threats caused by Phishing attacks and additionally, be able to identify the characteristics of this type of attack.

### **6.3.2 Specific Objectives**

For achieving the main objective this thesis is guided by the following specific objectives:

1. Identify characteristics of phishing attacks and how they are shown to users to trick them.
2. Determine unsupervised Machine Learning models and compare them for selecting the most suitable, for constructing the characterization model.
3. Implement an unsupervised Machine Learning model based on the previously selected algorithm.
4. Evaluate the performance of the model by execution times and performance metrics for refining the model and lowering the response times.
5. Understand the characteristics and behaviors of Phishing attacks with the help of the implemented unsupervised Machine Learning model, by studying the clusters generated by the model.

## **6.4 Limitations**

Alongside the development of this thesis, there are some limitations that are bound to the work:

1. Limited access to data: For developing this thesis the data used was found directly on the Internet, thus making the result not have the sufficient depth on analysis.
2. Time constraints: As this work represents an undergraduate research project, some aspects may have been rushed in order to obtain results as fast as possible due to the three-month deadline.

3. Deliverable Result: Due to the lack of monetary resources and advanced equipment, the whole project has been developed by an undergraduate with the tools at his disposal, including a personal computer thus the implemented model may not be fully optimized and its level of complexity may be high.
4. Extent: As this thesis work aims to develop an unsupervised model with one algorithm, the analysis done may not be highly detailed, for that reason it may prove to be limited as the final model is implemented with only one algorithm.

## 7 Theoretical Framework

### 7.1 Machine Learning

According to IBM, Machine Learning (ML) is a branch of Artificial Intelligence (AI) in which computers use algorithms that allow them to learn different patterns on training data for later making accurate and precise inferences of new input data. This understanding of the data enables the models to be able to take decisions without programmed instructions. [14]

With regard to ML it is one of the most important elements inside the AI world, this is because it represents the base for many AI systems at the moment, and a solid base for data analysis. Inside all of this systems the most common ones are the Large Language Models (LLM) and Generative AI, which can be found today as AI tools like ChatGPT or Gemini. The key to this is optimizing the performance of the model with a dataset that can represent the real world in the most accurate way possible, taking into account that, the model can generate an understanding of how the data is structured and composed, enabling it to elaborate different predictions on new data. In other words, a ML model generates an abstraction of the world, and because of that it will try to approach data to the most accurate inference possible with the help of different analyzed patterns and behaviors. [2][9]

It is important to notice that ML is directly linked with data science, the main reason for this is that it enables the analysis and synthesis of big amounts of data, for later on applying the learned elements into future data analysis. However for realizing more in-depth data analysis Deep Learning (DL) is used more frequently, as it allows a further understanding on the input data.[14]

#### 7.1.1 Deep Learning

Deep Learning (DL) is a subset of Machine Learning (ML) which centers on the work of deep artificial neural networks. In opposition to common ML, this subset is focused on the interaction of distributed "networks" composed of different mathematical operations which enables the possibility to study and analyze complex data. Due to the fact that DL is based on high data quantities, it requires huge computational resources for being able to process all of this data, thus it is commonly related to *Big Data* and *Graphics Processing Units (GPU)* processing.[14]

#### 7.1.2 How Does it Work?

Before all, it is important to state that ML is branch of AI this is mainly because the ML works with algorithms for learning the most implicit data patterns, thus

making it a more flexible, scalable and accessible AI system.[14]

Getting into detail, ML works by using mathematical logic and the identification of the most relevant features of data, this elements should be expressed in numeric terms and once they pass through the algorithm the input elements are mapped to an output. In the ML world the data is represented as vectors, this vectors have a numeric value for a specific characteristic and for cases when there is no numeric data, it must be converted to numeric so that the algorithm can elaborate the adequate mapping, so for instance when working with texts there is the option to use Bag of Words (BoW) or embeddings, as they map the text into a mathematical plane, giving each word a numeric value.[14]

Moreover, for training a model it is important to select the most suitable features, for doing so there has to be done a manual process of understanding and choosing the most meaningful elements to be used in the model. Once all this pre-processing is done and the data has passed through the algorithm it will generate different model parameters and give them a specific value or weight, by training with new features or new transformations the weight of these parameters may be subject to change thus increasing or decreasing the model accuracy. Additionally, depending on the input data and the analysis objectives there are different algorithms that can be applied for predicting or grouping data, the following sections will talk about the three main types of learning models that exists.[14]

### 7.1.3 Supervised Learning

Inside the Supervised Learning the task is either to predict a data value given some input data (regression) or to classify a data element based on the characteristics the element has (classification). Both of these should be done without sacrificing too much accuracy on the model, this due to the reason that a higher accuracy means a lower probability of making a wrong assumption, whereas a lower accuracy implies a higher probability of making mistakes. The accuracy is obtained after comparing the predicted result for the data entry, with the original value and "*correct*" output which is given as a label on the entry data.[14]

The main objective of a prediction model is to be able to reduce the loss value for a data entry, this loss value represents the difference between the original value and the predicted with the model. The name supervised comes after the fact that the labels for the true value are stated by an external actor or a human being, not exclusively done with machines.[14]

On the field of regression, the most common algorithms are linear regression and polynomial regression.

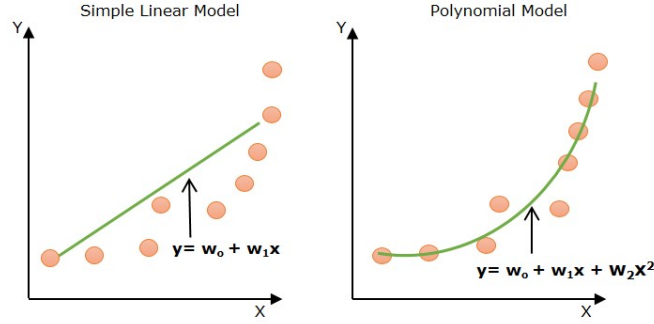


Figure 1: Linear and Polynomial Regression Algorithms [16]

In basic terms the regression approach searches to analyze the relationship between a dependent variable and many independent variables. Linear regression is used when the relation between the variables imply similar changes on both, in other words, if the independent variable increases or decreases the dependent has the same behavior respectively. Moreover, the polynomial regression searches to fit to a non linear data as it can adapt and generate a curve that fits towards most of the points. [15][16]

On the other hand, for classification two of the most common algorithms are Support Vector Machine (SVM) and Decision Tree.

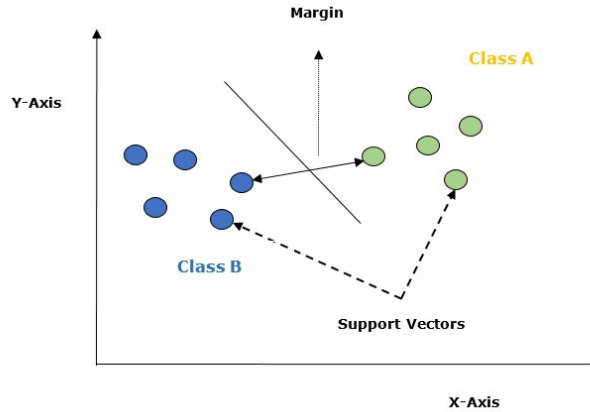


Figure 2: Support Vector Machine Algorithm [17]



Support Vector Machine aims to generate a plane in which it can separate data into different classes, the main objective of the generated hyperplane is to maximize the distance between the nearest elements among all the classes, in other words, the SVM aims to separate the data in groups so that these groups contain elements that are as different as possible. The plain also helps to generate the Support Vector which are the points that are more closer to the hyperplane. [17]

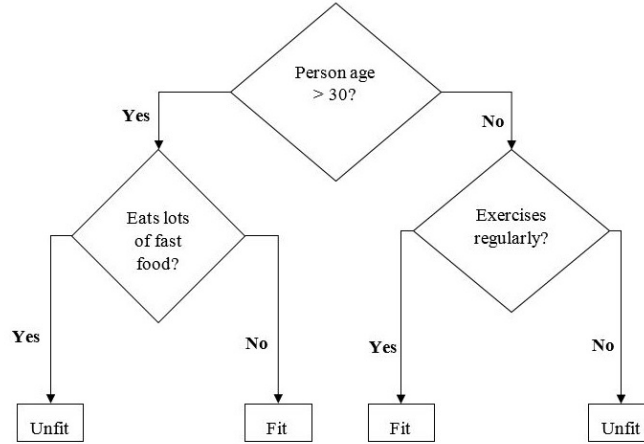


Figure 3: Decision Tree Algorithm [18]

In another case, decision tree is an algorithm which aims to separate data based on a set of rules, once the data enters the algorithm it tries to separate it on subsets giving a higher priority to the attributes with the most value. The aim of the algorithm is to split data until all the data inside a subset has the same target value. The final product the algorithm produces is a tree that represents the set of decision rules used to make predictions. The Tree above represents a case in which the data can be categorized into two classes, fit or unfit, and it also shows the set of rules used for making the categorization process. [18]

#### 7.1.4 Reinforced Learning

The main difference of this type of models is the fact that they are train through trial and error, making it the most similar to the human learning process. The main area of application for this are those such as robotics or video games, in which there is a huge number of possible solutions, thus defining a list of the solutions can be difficult. Instead of the classical objective of minimizing error, this models search to maximize the reward which is the feedback it receives after executing a task, this can be either positive or negative. This score can be calculated by a reward function or explicit rules, depending on the context in which the model is implemented. [14][19]

The main elements that build a reinforced learning models are the following:

- **State Space:** Contains all information related to the possible decisions that can be made.
- **Action Space:** Contains the decisions the model can perform.
- **Reward Signal:** The score an agent receives after each action.
- **Policy:** The thinking process that guides the agent to act depending on the rules.

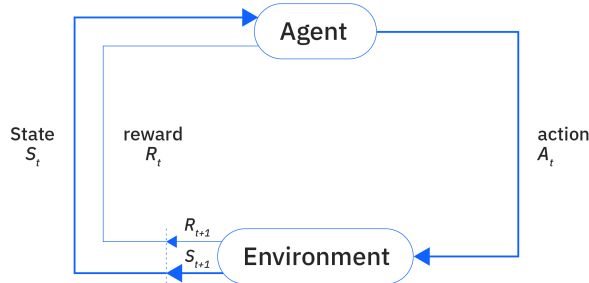


Figure 4: Reinforced Learning [19]

The agent is always looking to act for having the best score possible, thus it will try to perform every decision and evaluate the score it receives for it, by this it learns to avoid performing task which represent an insufficient score.

### 7.1.5 Unsupervised Learning

The main objective of Unsupervised Learning is to draw patterns, similarities or groupings from data with no labels. This type of models are useful when the labeling of data can not be done without a more in depth analysis, thus the lack of labels on the data makes it human free, therefore the name Unsupervised.[14]

Clustering is the most common task these models perform, in which they aim to group data based on the similarity between many data points, this proves to be very useful for client segmentation or fraud detection. On the other hand, Association aims to judge correlations between actions and conditions, thus they are adapted to work as recommendation engines. Finally, Dimensionality Reduction searches to lower the complexity of data, but at the same time preserve the more important features.[14]

This document will focus on Clustering thus it will be the main approach given to unsupervised learning. Because of that, the two most common clustering algorithms are K-Means and Hierarchical.

By its part, K-Means consists on the generation of random centroids for a given number of clusters and iteration over the centroids until these do not change anymore. The centroids change with every iteration so that the square distance of all the points to a centroid achieves its minimum, thus generating the grouping of different data points.[20]

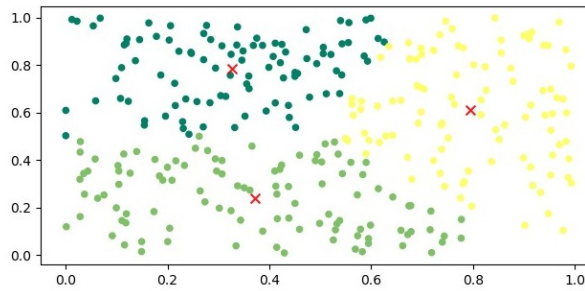


Figure 5: K-Means Algorithm [20]

On its contrary, Hierarchical clustering works on assuming data points belong to a specific cluster and afterwards the cluster can either separate into more groups (Divisive) or two or more clusters can group into one sole cluster (Agglomerative). The main difference with K-Means is the fact that it does not require a number of clusters, instead the number will generate after all the data points are assigned to a cluster. One tool that is useful to understand how this algorithm works is with the help of dendrograms, as it enables to see a hierarchy of the data points.[21]

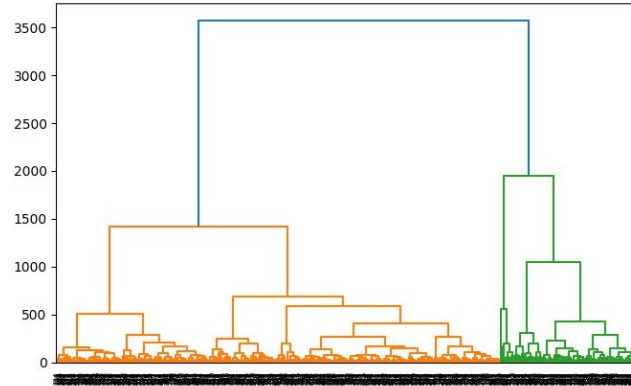


Figure 6: Cluster Dendrogram [21]

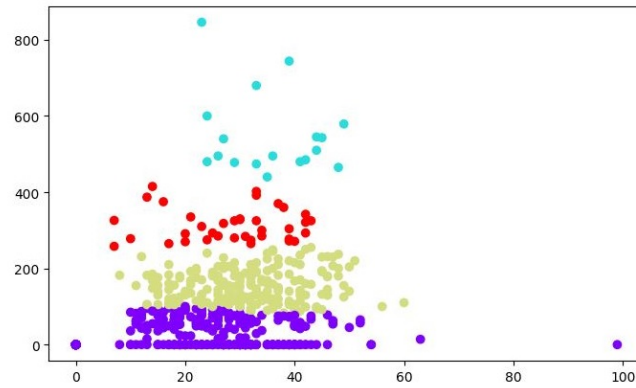


Figure 7: Hierarchical Clustering Algorithm [21]

It is important to notice that as this is the Machine Learning path that the work on this document will aim to achieve, there would be a more deeper analysis of Unsupervised Learning, including the algorithms that are aimed to use for the model.

## 7.2 Unsupervised Learning

As said before Unsupervised Learning is the use of machine learning algorithms for analyzing and clustering datasets with no labels on it, by discovering patterns and relations that are not so clear for human analysis. Unsupervised learning has three main stages that it follows during the analysis of data. For instance the first stage is the Interpretation, during this stage the data is studied for figuring out main characteristics that can be used for generating groups. The

second stage is the algorithm stage, in this stage an algorithm depending on the task type is applied in order to identify patterns in the input data. Finally the processing stage represents the moment in which the algorithm works over the data and thus it generates the different groups depending on the previously identified characteristics. [24][25]

It is important to notice that the main advantages of this type of learning is the freedom of working with no labeled data, thus the model can work with raw data saving time on cleaning and formatting it. As said before, this type of learning path helps to discover hidden groups to the human eye, it is useful specially when the input data is really big and the human brain is unable to process all this data and draw preliminary categories out of it. Finally, it is a good source for discovering anomalies, by its part algorithms like Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) are able to find outliers points on the datasets the model is working with. [24]

However, it is important to understand that the lack of data preparation can imply on the use of noisy data for the model development. In many cases the noise can distort the groups generated and will make it sometimes difficult to separate groups that look to much alike. Another problem is the fact that the clustering algorithms depend to much on assumptions, and in many cases the assumptions done may not fit into the data, making the groups overlap each other. Finally, algorithms like K-Means are sensitive to hyper-parameters and changing them even a small amount may cause the model results to vary significantly. [24]

Unsupervised Learning divides on three main classes such as clustering, association and dimensionality reduction, these are going to be explained on the following pages.

### **7.2.1 Clustering**

According to the previously said, this is the most common task when talking about Unsupervised Learning. The main objective of this activity is to perform data mining for drawing possible groups based on how similar are the data points inside of it. Clustering tasks divide on four main types, which would be explained next. [25]

### **7.2.2 Exclusive Clustering**

The main argument of exclusive clustering is providing each data point one and only one group, this because a point can only belong to a single cluster. This is sometimes referred to as hard clustering, because it forces points to decide for a cluster even though it may have characteristics of two or more groups. The main algorithm for this type of clustering is the K-Means algorithm as it forces points

to a cluster depending on the distance it has from each cluster centroid. The main aspect of this algorithm is the definition of a good  $K$  number of clusters, big numbers can imply in granularity but smaller numbers can end up causing generalization of data. [25]

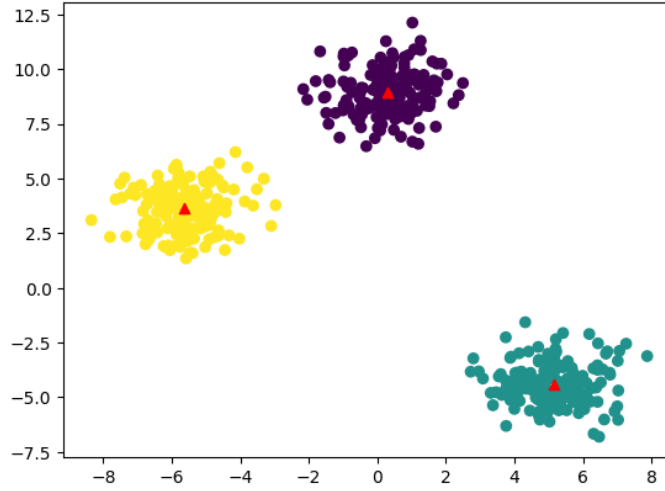


Figure 8: Exclusive Clustering [26]

### 7.2.3 Overlapping Clustering

On contrary to Exclusive Clustering, the Overlapping Clustering gives more freedom to the data points for the settlement of a group. Because of the opposite it has with hard clustering, it is denominated as soft clustering. In many cases the algorithms used for exclusive and overlapping share similarities, but in the end they differ on how strict they act among the data. For example, the most common algorithm is fuzzy K-Means, which has the same grouping logic as K-Means but not the strictness. [27]

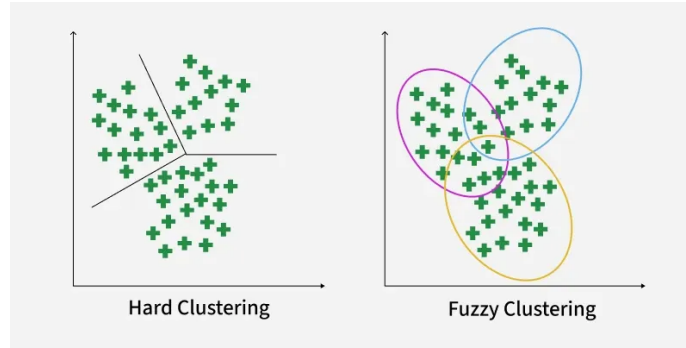


Figure 9: Exclusive vs Overlapping Clustering [27]

As the above figure shows, hard clustering restricts the freedom of a point to belong to different groups, whereas soft clustering allows points to be allocated in more than one group.

#### 7.2.4 Hierarchical Clustering

Moreover, Hierarchical Cluster Analysis (HCA) has two main types for grouping data being agglomerative and divisive. The first one for instance, isolates all data points into separate groups, then it aims to merge these groups into a main cluster depending on the similarities the smaller clusters have. It is also known as bottoms-up approach as it aims to build bigger clusters using smaller ones as a basis. Inside this type of clustering the most known algorithm and method is the Ward linkage, which states that cluster distances, calculated using Euclidean metric, are calculated from the increase of the sum squared once the clusters join. [25]

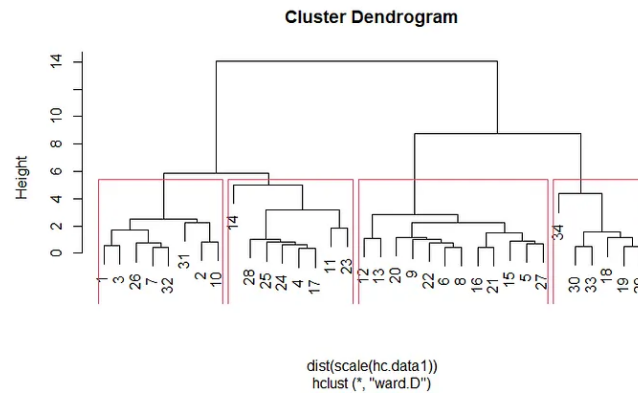


Figure 10: Agglomerative Clustering - Ward [28]

On the other hand divisive clustering does the opposite of the agglomerative method as it takes a top-down approach. In other words, this method starts with a global cluster and starts to break it down depending on how different are the data points inside the cluster, because of this it is rarely used. [25]

### 7.2.5 Probabilistic Clustering

Finally, probabilistic clustering works by the probability of a data point belonging to a particular distribution. The most common algorithm is the Gaussian Mixture, which works by determining the probability a data point has of belonging to either Normal or Gaussian probability distribution. In case the mean and variance of the data are specified, the distribution can be found almost instantly. [25]

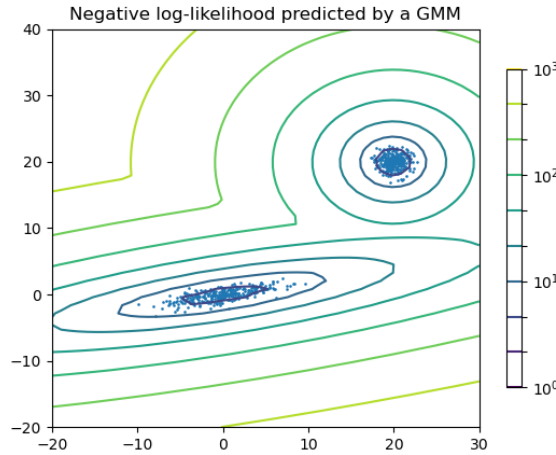


Figure 11: Probabilistic Clustering - Gaussian Mixture Models (GMM) [29]

### 7.2.6 Association

The main focus of association is the use of different rules in order to find relations of variables on a dataset. This branch is used for market basket analysis as it helps companies to understand how different products relate with each others. A very good example is the Amazon list of "Customers Who Bought This Item Also Bought". [25]

It is important to say that this type of task will not be analyzed in depth, as the main focus of this project is to implement a clustering task.



### 7.2.7 Dimensionality Reduction

Ultimately, dimensionality reduction aims to reduce the data inputs into an amount that can be manageable without affecting the integrity of the data, as in many cases the huge data amounts can affect the algorithms performance and also it may difficult the visualization of clusters. Some of the most common methods are Principal Component Analysis, which compresses datasets using a linear transformation. Also, Singular Value Decomposition aims to reduce a matrix  $A$  into three simpler matrices that preserve the following formula:

$$A = USVT \quad (1)$$

It is important to note that the produced matrices are  $U$  and  $V$  which are orthogonal and  $S$  which is a diagonal matrix. [25]

### 7.2.8 Scikit-Learn

Scikit-learn also know as Sklearn is an open source project and the most used Python library for Machine Learning ML, as it provides a wide range of tools for statistical data modeling and analysis. Inside this library there are plenty algorithms used for classification, regression, clustering and dimensionality reduction tasks, the algorithms are implemented with the help of NumPy, SciPy, Matplotlib and even more Python essential libraries.[34]

The main objective of this library is to simplify the realization of Machine Learning ML models as the brute implementation of them are bounded to an intensive error-prone task, which demands huge knowledge on statistics, algebra and calculus. Thus Sklearn comes as a solution and a viable option for implementing models without additional effort, and because of that it is going to be used as the main library for developing the clustering model of this project. [34]

### 7.2.9 Algorithms

Taking into account that there are numerous tasks unsupervised learning can cover, there are also different algorithms that can work for different objectives and depending on the data amount the work will be done with. This makes that choosing the most accurate algorithm is simple, on the following table it would be shown the comparison for the most common algorithms that are used for clustering implementations according to scikit learn. [30]

| Algorithm           | Scalability    | Purpose                                 | Metric Used                     |
|---------------------|----------------|-----------------------------------------|---------------------------------|
| K-Means             | Large Samples  | Flat Geometry, Small Cluster Amount     | Points Distance                 |
| Spectral Clustering | Medium Samples | Non-Flat Geometry, Small Cluster Amount | Graphs Distance                 |
| HDBSCAN             | Large Samples  | Non-Flat Geometry, Outlier Removal      | Distance Between Nearest Points |
| Gaussian Mixtures   | Not Scalable   | Flat Geometry, Density Estimation       | Distances To Centers            |

Table 1: Clustering Algorithms Comparison

In general, and after checking the four main algorithms used for clustering tasks, this project is gonna be based on K-Means algorithm as it is the one that adapts better to the projects needs and context, this however will be explained more in detail later.

### 7.3 Cybercrimes

In simple terms Cybercrimes consists on all the illegal actions that are committed by using different devices connected to the Internet, these actions can go from the spread of a malware file, up to identity thief. The main actor that performs this actions is someone with access to a device connected to the internet, and the main goal this person has is to harm others. The three most common categories are the attacks on individuals, which consists on attacks to normal internet users, in these identity thief is the most common one. Also, there are attacks on property or systems, these consists on focused attacks to affect business or technological systems, DDoS and malware installing are the most frequents ones. Finally, the attacks on government consist on the cyber terrorism and propaganda spreading, the objective of this type of attack is to target governmental sites, as they have valuable information. [22][23]

Additionally, it is important to focus on the aspect that Cybercrimes does not need a physical and close action, it can be performed by people located on a vast distance. Thus this makes it difficult to handle them locally, and because of this the topic has come to be a global matter and the dealing of it requires the international cooperation. The main reason for this type of actions is the fact of having computers and the internet as our central element of life now on days, and also due to the matter of having a digital identity. This gives people a virtual representation that can be easily attacked as it is very fragile because unlike a physical identity, it is difficult to prevent and respond to attacks in many cases, as this digital identities in many cases are not localizable. [23]

Due to the huge variety of attacks that exists on the present, it would be difficult to cover them all, in exchange the following work will focus only on Phishing attacks and on the next part it would be explained what is Phishing and how does it work.

### 7.3.1 Phishing

In general terms, Phishing is a type of cyber-attack in which the victim receives electronic mails, text messages, calls or even fraudulent web sites, so that he can be tricked to share his personal and confidential data, download malware directly or even create breaches for further cyber-attacks. [11]

According to Kosinski (2024), Phishing attacks are based on Social Engineering, this makes them different from common attacks which target networks and resources directly, as this attacks are more user focused. The main objective of this attack is to force user mistakes, as the attacker uses the identity of someone the victim trusts such as a co-worker, brand or even family member. Once he enables contact with the target, they send messages in which they tell the user to open an external link, transfer money or even download files. As the user is unable to identify the attacker, he follows the instructions and ends up falling into the attackers trap.

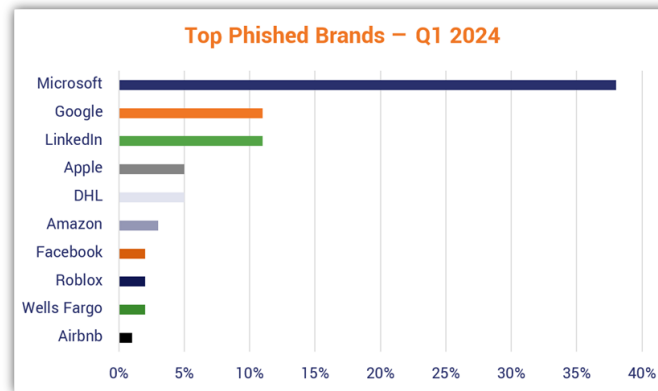


Figure 12: Most Phished Brands on Q1 2024 [13]

The previous image shows how the most common imitated brands are the most famous ones on the digital world, brands like Microsoft and Google are commonly known for their big influence and the image they have on their users. However, being the most Phished ones is a problem because users tend to believe the content they receive from them, and having almost 40% of the content declared as fake come supposedly from Microsoft is concerning.

Additionally, Kosinski (2024) states that the main concerns about falling into the attackers hands are that they have multiple ways to harm people, from downloading a file that can contain a virus or a ransomware, to making them access an external website that is an exact replica of the original site, and in this site they can steal all the information the user enters.

Furthermore IBM study "Cost of a Data Breach" shows that Phishing is the most efficient cyber-attack used by cybercriminals, as it represents the 15% of the total vulnerabilities people fall into arising the costs to an approximate of 4.88 million dollars by organization.

Going back into Kosinski (2024), he states that Phishing is one of the most threatening attacks on the Internet, as it instead of making use of technological vulnerabilities it makes uses of the people and often tricks people with low digital knowledge. The issue are the users the criminals focus, these users are people that have authorized access to information, services or systems and once they trick these people they simplify the job of stealing as they do not have to force their entry onto the systems breaking all the security tools the system has implemented. One of the most recent cases of Phishing was the one in which Russian hackers used fake e-mails for resetting the password of Hillary's Clinton mail, and once they tricked her they ended up stealing thousands of mails related to her presidential campaign of 2016. The main reason why these mails were not detected was that the informatics assistance service saw the fake mails as authentic ones, therefore they were not discarded.

Moving on, the term Phishing comes from an analogy of the word fishing in which the fishermen use different bait for hooking fishes, therefore Phishing makes reference to how the attackers can bait people into making them believe that what they are receiving is true, and by that way they can make them evoke different emotions that lure them. [12]

Inside the whole Phishing concept there are different types, which are explained below [11]:

### **Masive E-mail Phishing**

On this type of Phishing, attackers send huge numbers of e-mails using the identity of a big company, copying their logo and domains, so that a fraction of the people proceed accordingly with the mail content. The majority of this attacks are registered on special dates, for example, Amazon Phishing attacks arise to their peak on Prime Day.

### **Spear Phishing**

This attacks are targeted to a specific person, commonly the objectives are people with access to critical information or even the financial accounts for lurking money. Inside these exist the **Whale Phishing** which target high

power directives or objectives.

### **Business Email Compromise**

Business Email Compromise (BEC) are a specific type of Spear Phishing which attempt to try to steal monetary resources or valuable data from organizations. It has two main forms:

#### **CEO Fraud**

The scammer gains access to an executive account and communicates with the lower-level employees making them transfer funds into fraudulent accounts.

#### **Email Account Compromise**

In Email Account Compromise (EAC) the fraudster obtains access to a low-level employee and uses the account to contact vendors or instruct other employees to commit fraudulent payments.

### **Smishing**

Smishing is a type of Phishing that is committed directly by text messages or SMS, usually the messages the victims receive are about free gifts and ask for credit cards data.

### **Vishing**

This type of Phishing takes place on phone calls, according to the APWG this attacks increased by 260% from 2022 to 2023. The main reason for this is the increase of the Voice over IP (VoIP). The main reason why people are tricked is the fact that scammers use local phone numbers, so the victims does not suspect at all.

### **Social Networking**

Some scammers resort to social media with advance messaging like Facebook, LinkedIn or X. The main resource they use is the login process as they hope to acquire login credentials.

Additionally, AI has contributed to the development of Phishing as it helps to generate personalized content with no human error, so the victims do not recognize it is a fake message. Also, it also helps the criminals to accelerate their operations as they can generate thousands of mails and content to spread within a single prompt. [11]

## **7.4 Social Engineering**

On the another hand, Social Engineering is the act in which people are manipulated to share confidential information, download Malware or visit fake websites, so that digital criminals have an easier job obtaining this kind of information. This is effective as it works by psychological manipulation with the objective of

making people commit errors or show any type of weakness so that there can be a breach that can be used as an exploit. Because of this, the user is the one who ends up revealing all his personal information to a criminal and later his identity would be stolen by these people, so they can either trick someone or start more cyber-attacks. But the main reason why this is commonly used among criminals is the fact that it reduces the amount of effort they must put into accessing systems and private data, because by obtaining authorized credentials or logins, they do not have to break firewalls, anti-viruses and other Cybersecurity controls, they just need to access the site using the right information and they can steal easily. [4]

#### 7.4.1 Social Engineering on Phishing

It is important to notice that Social Engineering and Phishing are related concepts as Phishing attacks are a particular type of Social Engineering attack. More in depth, Phishing is the Social Engineering enforced through messaging platforms for sending fake links, malicious files, or other threatening content for the user, so that the user can bond with the attacker. [32]

As mentioned before, social engineering aims to trick a user by provoking emotions on it, however it is difficult to prevent this as the user can feel different emotions depending on the content inside the phishing the user is receiving. The most common message contents are the ones that state that you account is about to be suspended, due to expired payment or terms of service violations. Also, there are fake job scams which supposedly offer a job opportunity with high payment and nearly no skills, this lures people as their desire for easy money is being fed. Finally, another case is the fake orders emails, which contain information about non existent online orders, this emails pressure the recipients to act quickly and follow the attackers instructions. [33]

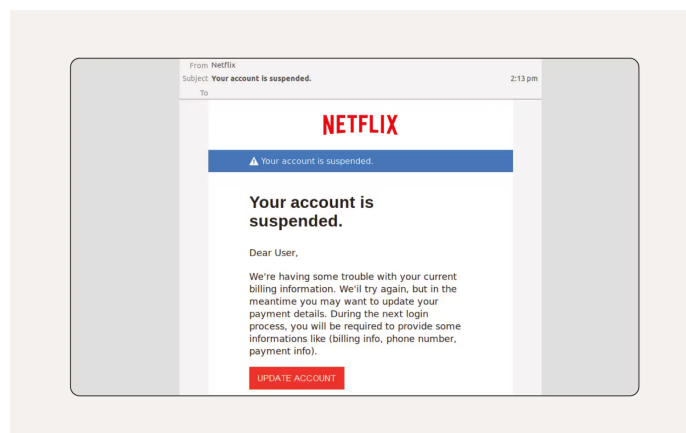


Figure 13: Example of a Phishing Email [33]

## 8 Metodology

### 8.1 Approach

In order to combat the Phishing attacks and propose a solution, the following study and project implementation aims to perform a qualitative approach to Phishing emails by generating and training an unsupervised machine learning model centralized on a clustering task. Thus, the focus will be the implementation of a clustering algorithm for extracting the main patterns after grouping different emails based on their sender, receiver, subject, and content.

### 8.2 Research Design

The work makes use of computational resources for simulating an email characterization with the help of clustering algorithms. Once the emails have been processed on the model, the algorithm will assign them to a group for later or extracting the most important attributes of the groups.

### 8.3 Data Collection

For performing the study, data has been collected from Kaggle which is an online platform centralized on data science and machine learning, that has been operating since 2010. The main reason for using Kaggle is the fact that it is home to huge amounts of resources like datasets, code notebooks and tutorials that are used for working on real-world problems with the help of a community build up by both enthusiasts and experts. [31]

Inside Kaggle the dataset for working was found under the name of Phishing Email Dataset. Data set would be included in the Appendices for this work.

## 9 Development & Proposal

### 9.1 Preliminary Model

In order to get a detailed idea of the process that will define this thesis project the first approach was done with a preliminary model which will use a data set extracted from Kaggle about Phishing emails. The objective was to elaborate different models for obtaining the most accurate one for trying with a wider data sample and certain optimizations.

#### 9.1.1 Data Structure

To begin with, for developing a preliminary model, it was essential to select a data set with a small amount of data to make a test of the idea proposed to implement. For that reason, the data set was taken out of Kaggle and can be accessed at the Appendices.

Inside Kaggle, the data is found as a CSV file with 7 different columns, which are:

- **Sender:** Name of the email sender
- **Receiver:** Name of the email receiver
- **Date:** Date and time the email was created
- **Subject:** Subject the email was created with
- **Body:** Content of the email
- **Label:** Classification
- **URLs:** Whether the email contains URLs or not

However, not all the columns are needed for the model; therefore, the next step was to check the data and prepare it for the model.

#### 9.1.2 Data Preparation & Preliminary Results

The main objective for preparing the data was to delete all unnecessary elements found inside the text, the objective was to extract the main elements of the email and keep it as plain as possible. For making that possible, the columns of date and label were eliminated from the data as they do not provide additional value to the email, and because we are trying to categorize and there is no need for having labeled data.

The next step was to clean the text inside the emails, for that the decision of removing American Standard Code for Information Interchange (ASCII) characters that could imply incoherence on the email content, latter for avoiding



repetition of the same word the whole text was passed down to lower case, this was done for avoiding interpreting words like "Account" and "account" as different words. Later the punctuation elements were removed due to their lack of meaning inside a text, after that the numbers were cleaned and transformed into words, by that way they can be conserved and at the same time give more information and meaning to the text. The next step was to remove the stop words such as articles and connectors, due to their lack of meaning and high frequency inside a text, leaving them would only affect the model performance. Afterwards, the clean text was joint together on a single sentence, which will go to the text interpreters on the next step. All of this was done with the help of the Python libraries spacy, nltk and num2words.

The next part consisted on working with two different forms for interpreting texts, which are the BoW and the text Embeddings. For the first one, the main objective is to tokenize a text and generate a matrix which represents whether or not a word is present in a sentence, however the results obtained by that method where not good at all, as it is not possible to see the groups in clear ways.



Figure 14: Results Obtained with the Bag of Words Strategy

Because of that it was essential to test a different strategy, in this case it was decided to implement text embeddings, as it helps to give each text a numeric value in a vector space, depending on the elements inside of it. With this method the clusters generated were easier to see and distinguish.

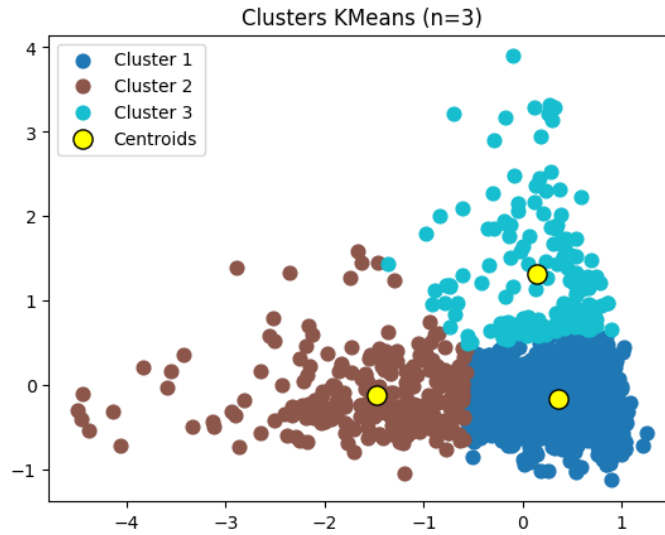


Figure 15: Results Obtained with the Text Embeddings Strategy

The previous graphs are the ones obtained from the implementation of the K-Means algorithm, which turned out to be the best option of the five previously selected algorithms. However, the results for the other four algorithms can be visualized on the repository which link is included on the appendices.

However, the next table shows the results for the five different algorithms that were tested on the preliminary model:

| Algorithm           | Execution Time | Number of Clusters | Silhouette Score | Cluster Distribution                                                        |
|---------------------|----------------|--------------------|------------------|-----------------------------------------------------------------------------|
| K-Means             | 135.339 sec    | 3                  | 0.28735          | 71.23% / 10.5%<br>/ 18.27%                                                  |
| Spectral Clustering | 234.809 sec    | 2                  | 0.28626          | 12.54% /<br>87.46%                                                          |
| HDBSCAN             | 134.383 sec    | 2                  | -0.33216         | 1.43% / 98.23%<br>/ 0.34%                                                   |
| DBSCAN              | 138.049 sec    | 7                  | -0.26450         | 29.93% /<br>67.28% / 0.41%<br>/ 0.34% / 0.20%<br>/ 0.82% /<br>0.41% / 0.61% |
| GMM                 | 142.777 sec    | 4                  | 0.12411          | 46.97% /<br>25.77% /<br>14.11% /<br>13.16%                                  |

Table 2: Clustering Preliminary Notebooks Comparison

Something important to notice here is that HDBSCAN and DBSCAN both generate a cluster of Outliers (Cluster  $-1$ ), thus this is shown in the first percentage of cluster elements.

### 9.1.3 First Conclusions

In the end, the best algorithm for generating the final model is the K-Means algorithm due to its fast and effective work, as well as the generation of separated visual groups, thanks to the high Silhouette Score it achieves. Additional to that, the text embeddings strategy is the best one for transforming text into processable inputs, as it gives each text a different value depending on the content. Therefore for the final model it is important to work with the K-Means algorithm and text embeddings strategy.

## 9.2 Final Model

After working with the preliminary model, it was decided to take it as a base for developing the final model so that the process of ML was clear from the start and the steps to develop would generate significant results which could be later on optimized or modified. For that reason the model would work using text embeddings as the main strategy for preparing the data, and K-Means as the algorithm of election due to its capacity of performing well with great data amounts.

### 9.2.1 Data Structure

The data for this model is taken from Kaggle and can be checked on the Appendices, the file by it self conserves the same structure the preliminary data had, but this new CSV files contains more data on emails.

Therefore the file that is going to be used for the final model contains the following columns:

- **Sender:** Name of the email sender
- **Receiver:** Name of the email receiver
- **Date:** Date and time the email was created
- **Subject:** Subject the email was created with
- **Body:** Content of the email
- **Label:** Classification
- **URLs:** Wether the email contains URLs or not

A difference spotted with this new file from the previous file is the fact that the column label now contains 0 and 1 as values, meaning 0 the emails that are not classified as Phishing, whereas 1 represents the emails that have been classified as a Phishing email, because of this it is important to perform an additional step on the data preparation for working only with the emails that have a value of 1 on their label, as they represent the Phishing emails of the dataset.

### 9.2.2 Data Understanding

The first step was to understand the data that will be used for the final model, thus it was important to understand the structure it has, look out for any duplicates or missing values. But it was also important to generate a brief summary of the contents inside the emails, this was done with the help of a Word Cloud that will help to visualize the most common words inside the emails, the generated Word Cloud is the following:



to be the most efficient algorithm on the preliminary tests. For the number of clusters three is going to be the best starting option because it adjust perfectly with the three previous groups inferred after elaborating the first word cloud. Afterwards once the results are obtained, there is a process of analysis and understanding of the produced outcome, additionally, the results of the modeling will be discussed on the next section.

## 10 Results

### 10.1 Model

#### 10.1.1 Clusters

After realizing the data preparation and the modeling step with three clusters ( $k = 3$ ) the outcome produced shows that the cluster distribution is done almost equally. This can be said because it is possible to differentiate the three generated clusters with the data, and it is possible to see that the elements of each clusters does not overlap each others, additionally, the amount of elements present by cluster is almost identical as the clusters contain almost the same amount of elements and there is no cluster who groups more than 50% of the data.

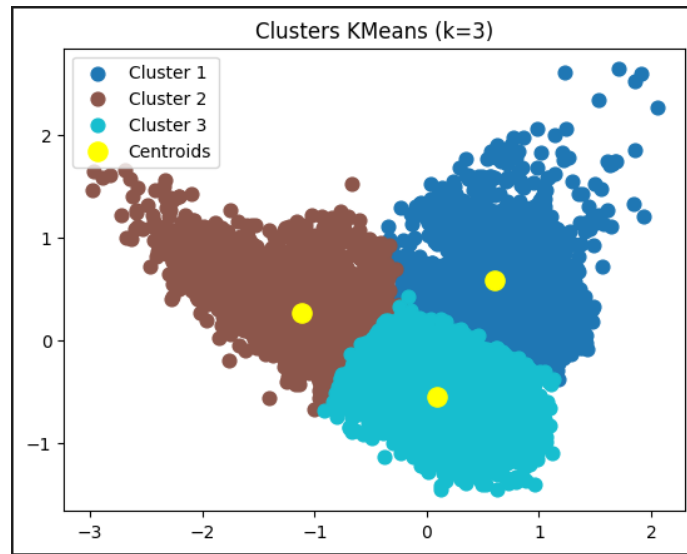


Figure 17: Clusters Generated With K-Means

As shown above it is possible to identify each of the clusters as there are no data points that are overlapping each other, there are some points that are as away as possible from each others, this can be said to be the group representatives as they can only belong to a specific cluster, this points are the ones that are on the edges of each cluster. However, there are also some points that can belong to any of the three generated groups, this is due to the fact that the clusters are not full apart from each other, but they are close by.

Now, it is time to talk about the element partition for each group.

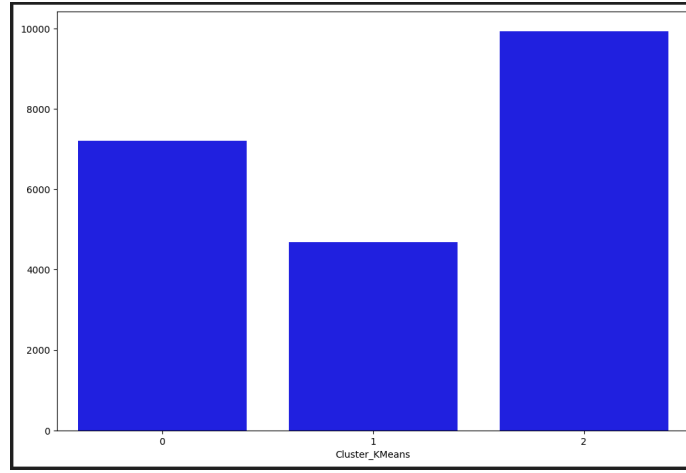


Figure 18: Clusters Members

| Cluster ID | Number of Elements | % Of Data |
|------------|--------------------|-----------|
| 0          | 7207               | 33.02%    |
| 1          | 4683               | 21.46%    |
| 2          | 9937               | 45.53%    |

Table 3: Clustering Data Partition

From this it is possible to say that the clusters generated are partitioned in a proper way as the percentage of data difference between clusters is of approximately 12% which represent around 2600 emails, thus this implies that the generated clusters separate data in an almost uniform way as there are no clusters that show a completely dominance of the data points, unlike the results seen on the preliminary notebooks, the clusters in here have a sufficient amount of elements to be considered independent clusters.

### 10.1.2 K-Means Metrics

The following part will be focused on the analysis of the Inertia and Silhouette metrics obtained for the elaborated model.

#### Inertia Score

Inertia score represents how compact the generated clusters are. It works out by calculating the sum of the square distances of the cluster samples to their respective center, a lower inertia score is the best option as it represents more compact groups. Additionally, to that the constant iteration of K-Means for different  $k$  values allows the study of the optimal number of groups for a model, this is known as the elbow method. With this method, the main objective is to



check for which  $k$  value and further, the inertia decreases slowly, this is better shown graphically as it allows to see for an elbow in the graph of  $k$  values and the inertia score. [35][37]

The following graph shows the inertia graph (elbow method) obtained for the proposed method:

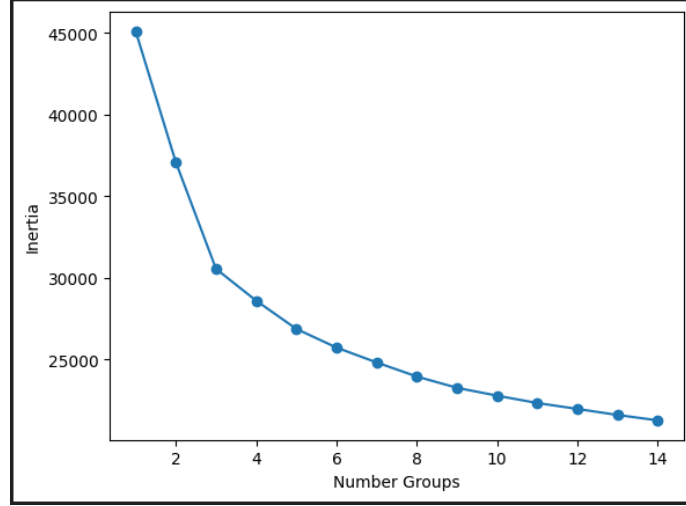


Figure 19: Elbow Method

From the graph it is possible to see that the break point on the  $k$  values is  $k = 3$ , therefore the Inertia Scores establishes that the optimal number of clusters the model should consider is 3, as the Inertia barely changes afterwards and incrementing the number of clusters can imply a reduction on the Silhouette Coefficient, which would be discussed on the following section.

### Silhouette Coefficient

Just like the Inertia Score, the Silhouette Coefficient is a metric which allows to understand how far apart are all the groups from each other. The mathematics behind this metric is the calculation of the distance between a sample and the closest cluster for which this sample does not belong, by its part, the coefficient makes reference to a global sample distance between the other clusters and the sample point. Its range is between  $-1$  and  $1$ , which means values closer to  $1$  mean that clusters are well defined and far apart from each others, values near  $0$  mean that clusters are overlapping and values near  $-1$  mean that the samples have been placed on the wrong group. [36]

Taking into account the previous information, the following figure represents the Silhouette Coefficient obtained.

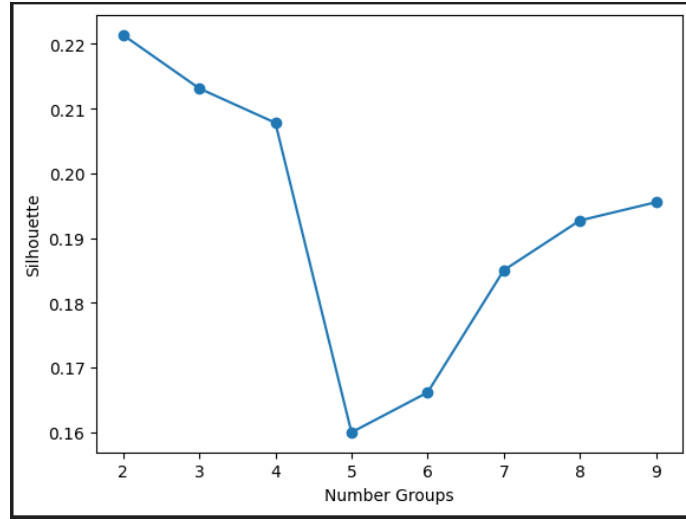


Figure 20: Silhouette Coefficients for different  $k$  values

The ideal value of the silhouette is for it to be the highest possible, but for the model elaborated it is when  $k = 2$ . However, the Elbow Method tells that  $k = 3$ . So which one should the model consider as  $k$ ? Well for answering this it is important to take into account both metrics for both values and understand the tradeoff of choosing one as a base for the  $k$  value, for making that the following table will show an approximation of the values for both  $k$  values.

| <b>k</b>            | <b>Inertia</b> | <b>Silhouette</b> |
|---------------------|----------------|-------------------|
| 2                   | 37000          | 0.221%            |
| 3                   | 30500          | 0.213%            |
| <i>k</i> difference |                |                   |
| 3-2                 | -65000         | 0.08%             |
| 2-3                 | 65000          | -0.08%            |

Table 4:  $k$  values metric comparison

From the table it is possible to see that choosing  $k = 2$ , implies on sacrificing a huge value of Inertia, therefore the groups may not be as compact as they are with  $k = 3$ , but choosing  $k = 3$  means that the sacrifice is of 0.08 of Silhouette Coefficient which along does not imply on a big change on the cluster separation, taking into account the data is too close so that the Silhouette score will not change drastically. Therefore after looking into the Inertia (Elbow Method) and Silhouette Coefficient, the best option is to remain with  $k = 3$ .

### Groups Silhouettes

Taking into account that the main objective for a model is to obtain the closest

value to 1, it is important to note that each group should have elements which surpass the silhouette average score. This can be seen easily on the following graph:

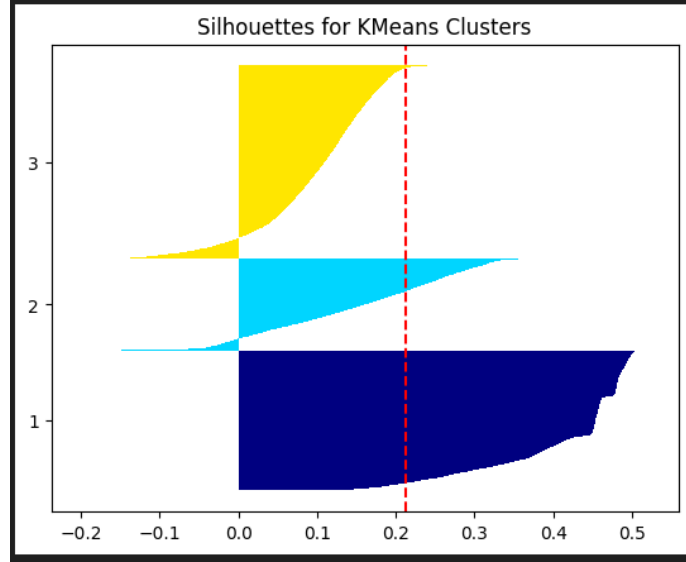


Figure 21: Silhouettes for the  $k = 3$  groups

The above image shows how the groups are well defined as they contain elements which Silhouette score are above the average Silhouette Coefficient. However, it is important to note that there are some elements which have a negative silhouette score, this shows that there is a percentage of elements of these groups that are located on a wrong spot. All of this is caused because the model is working with text embeddings, and taking into account that the texts have similar contents means they are located on a near position inside the vector space. In other words, the huge similarity between texts make them that they can belong to more than one cluster, but while working with hard clustering the model may assign some data points to the wrong group.

### 10.1.3 Cluster Characterization

After the elaboration of the model it is possible to draw the following three clusters:

#### Cluster #1 - News Alert & Social Content

This group is conformed by the emails with the fewest amount of external URLs, in this group 57.7% of emails contained URLs, additionally, it is represented by the emails with the second highest amount of elements and it contains the emails with the most constant length with it been of 250 words per email.

When talking about content, the email appears to have words like CNN, news, videos and alert, which represent the senders trying to trick the user into believing the email comes from an authorized news supplier. This group represents the emails with the biggest number of words on the content, with an average of 126.09 words per email body.

### **Cluster #2 - Replica Goods**

For this group the most common words refer to the user as a client and talk about replica watches and brand watches. This group represents the replica goods that the user receives advertising spam. Additionally to that, the group has a total of 64.1% emails with URLs and the mean length for the body is of 35 words per email, moreover this group contains the most emails with different sizes, unlike the previous group in which there is a constant of 250 words per email, this group has emails with 25, 50, 100, 150, 200, 250 and even 300 words, making it the group with the more inequality on the email lengths.

### **Cluster #3 - Pharmaceuticals & Dating**

The third group is the group that contains the emails with the majority of URLs as 77.7% of emails contain an URL. It is also the group with the smallest email sizes as the average of the emails size is of 26.56 words per email. In this group the main words are bonded to the pharmaceuticals a person can use to increase the size of its private parts, as well as the use of drugs and medicine for supplements and pleasure.

For more detailed views on the results, consult the Repository for the project found on the Appendices (see A) for this work.

However, it is important to look at the scores obtained for the model and understand the following: Due to the low Inertia and Silhouette scores, even though the groups are defined they are still so close to each other, thus this can imply on the mixture of some groups. The main reason for saying this is the fact that the groups are very close to each other means that the model needs further work for increasing these scores and for being able to avoid elements separation into no desired clusters. However, these results are generated due to the text meaning and the fact that many emails share similar content.

## **10.2 User Interface**

For presenting the model and making the users able to interact with it, it was decided to implement a simple and easy to use interface so that the user can input one email and once he submits it, the model answer will be shown.

### Email Cluster Assignment

Sender  
Daily Top 10 <Karmandeep-opengev@universahet.psi.br>

Receiver  
user2.9@gvc.ceas-challenge.cc

Subject  
CNN.com Daily Top 10

Body  

```

>===== >THE DAILY TOP 10 >from CNN.com >Top videos and stories as of Aug 1, 2008
3:58 PM EDT >===== TOP 10 VIDEOS 1. MONTAUK MONSTER
http://www.cnn.com/video/partners/email/index.html?url=/video/us/2008/07/31/moos.montauk.monster.cnn.is.it.a.deer.dog?is.it.a.turtle?is.it.the.Montauk.Monster?CNN's
Jeanne Moos asks "what is this thing?" 2. RACY PHOTOS OF TODDLER'S MOM http://www.cnn.com/video/partners/email/index.html?
url=/video/crime/2008/07/31/ng.racy.photos.cnn 3. NEWS OF THE ABSURD EPISODE 54 http://www.cnn.com/video/partners/email/index.html?
url=/video/podcasts/absurd/site/2008/08/01/nota.episode.54.cnn 4. POLICE BEATING DISPUTE http://www.cnn.com/video/partners/email/index.html?

```

URL  
1

1 For URL presence, 0 For URL absence

**Submit**

**Cluster 0 - News Alert & Social Content**

Words like CNN, news, top and network are in charge of talking about the news and the information work.

Figure 22: User Interface Elaborated for Presenting the Model

The main aspect of the interface is the fact that it uses the same data columns that the K-Means model used for training. Inside of the interface the user will need to input the sender, receiver, subject, body and the presence for an URL on the interface, once he has done that he would be able to submit the data and depending on the inserted data the model would return the cluster to which the element belongs to.

In this case the interface is using a fake email of CNN and because of that it assigns it to the cluster 0 which represents the News Alerts & Social Content. However, for avoiding empty data the interface has restricted the conditions of the input so that the user enters a valid content otherwise the system will not process the inputted data. By that way the interface grants the model that no empty data is gonna be send to it.

### Email Cluster Assignment

Sender  
Daily Top 10 <Karmandeep-opengev@universahet.psi.br>

Receiver  
user2.9@gvc.ceas-challenge.cc

Subject  
1

Body  

Invalid Input

```

>===== >THE DAILY TOP 10 >from CNN.com >Top videos and stories as of Aug 1, 2008
3:58 PM EDT >===== TOP 10 VIDEOS 1. MONTAUK MONSTER
http://www.cnn.com/video/partners/email/index.html?url=/video/us/2008/07/31/moos.montauk.monster.cnn.is.it.a.deer.dog?is.it.a.turtle?is.it.the.Montauk.Monster?CNN's
Jeanne Moos asks "what is this thing?" 2. RACY PHOTOS OF TODDLER'S MOM http://www.cnn.com/video/partners/email/index.html?
url=/video/crime/2008/07/31/ng.racy.photos.cnn 3. NEWS OF THE ABSURD EPISODE 54 http://www.cnn.com/video/partners/email/index.html?
url=/video/podcasts/absurd/site/2008/08/01/nota.episode.54.cnn 4. POLICE BEATING DISPUTE http://www.cnn.com/video/partners/email/index.html?

```

URL  
1

1 For URL presence, 0 For URL absence

**Submit**

Figure 23: User Interface Limitations for Empty Data

## 11 Discussion

Before continuing it is important to remember the main objective for this thesis work, which was the implementation of a machine learning model for detecting possible threats that can be caused by Phishing attacks, and additional to that be able to identify characteristics of this type of attacks.

### 11.1 Phishing Attacks Characteristics

From the implemented model it is possible to state that the Phishing messages aim to trick the user to believe that what they are reading is true. In many cases they use URLs so that the user desperately clicks it before even thinking if it is a good idea or not. Also, from the generated clusters it was possible to understand the following:

1. The content of the emails in some cases aim to copy and supplant trust authors and/or entities thus making the receiver believe the content of the email is true, in this case CNN was shown as the supplanted entity.
2. Some content inside the email focuses on offering goods or products that can be of the interest for the user. In this case the enhancement drugs and herbal supplements aim to lock the user attention so they are intrigued to discover more of these type of products.

### 11.2 Machine Learning Models

Before beginning to develop the final model and due to the size of the training data for the final model, it was decided to generate a preliminary model using different algorithms for looking at their results and metrics so the final model would be based on the best one, and by that the final model would be implemented having some base process, instead of going into discovery of the best method. After the realization of five preliminary models, with K-Means, HDBSCAN, DBSCAN, Spectral Clustering and GMM it was decided that K-Means represented the better results as the scores obtained both for time and silhouette proved to be the best ones of all the five models.

### 11.3 K-Means Machine Learning Model

After the realization of the five preliminary models and the analysis of the best one to implement it was decided that K-Means proved to be the best one out there. Afterwards, the final model was implemented using K-Means and once the model was generated it was important to take the metrics of Inertia and Silhouette to check the conformation of the three different groups. It also helped to implement a graphical visualization of both the elements in the groups and the groups forms, with that in mind it was possible to assimilate that the generated groups have a defined form.

## 11.4 Model Metrics

Once the metrics were measured it was important to analyze them and with that in mind it was established that even though the model generated the respective groups and they were differentiable in terms of word contents inside of them, the model still needs some refinement so that the metrics can improve their scores. However, the elaborated model allows to understand and interpret the different characteristics the Phishing emails contain.

## 11.5 Cluster Analysis

Regarding the three generated clusters it was possible to infer that Phishing emails contain external URLs in most of the cases, all the clusters have more than 50% of emails containing external URLs. Also, it is important to note that all the clusters contain senders which are clear replications from well known entities, some of these are google and CNN between others. Moreover, the emails use words for focusing the user attention on them, some of these common words are: alert, cheap, effective, help, information, online, recommended and top. Finally, it is important to note that these emails have considerable lengths to appear to be real, the user with that in mind would read the whole email and due to the length of it, the user will believe even more that what it is reading is true.

## 12 Conclusions

To conclude, the proposed work achieves the established main objective of been able to implement a machine learning model for detecting possible Phishing threats and elaborate a characterization of Phishing attacks. With that in mind it is important to state that the model achieves this objective as it works strictly on Phishing emails as a base for understanding the characteristics that these have in common. Furthermore, it is important to note that the elaborated training process allows the model to separate the whole email dataset into three clusters which cover three of the most common topics found inside Phishing content, which are fake news, replica products and pharmaceuticals related to pleasure.

Moreover, with this work it is possible to understand how does the Phishing emails are structured and how many of these aim to replicate famous entities and make the user believe that what they are reading is true, also it is important to note that in many cases this emails are received with an external URL, thus this makes the user hesitate not to click it as the content found on the email usually refers to this external link. Overall, the model can be used to teach people who are unaware of Phishing and its consequences, how does a Phishing message looks and what do they aim to do to the user. In other words, the elaborated model can be considered as a model for raising awareness about Phishing, due to the huge number of cases and message varieties that can be found now on days. And with the help of the model users can learn how do some Phishing emails look, this includes all the emails related to fake news, pharmaceuticals and replica products.

However, as mentioned before it is important to note that this work is subject to limitations and this limitations imply on a loss of efficiency and effectivity when assigning a cluster to an email. Because of that, there are some cases in which the model confuses whether or not to assign a data point into a cluster, and in some cases assigns it to a wrong group. All of this is understood when looking at the metrics and understanding that the model aims to generate the respective clusters, however it would need a further development for been able to separate the clusters completely and avoid mismatch elements between clusters. Even though this is an issue found, it is not a critical issue as all the data used for the elaboration corresponds to Phishing emails, and the content found inside this emails is partially related, thus it may categorize wrong the type of Phishing it is but in the end it is still Phishing. Still, it is important to notice that in case no Phishing emails were used it would be a critical issue as it may be assigning no Phishing emails to a Phishing cluster.

In the end, the developed work achieves the main objective of creating and implementing an unsupervised machine learning model with the K-Means algorithm for been able to generate the respective clusters and afterwards understand the characteristics of each of these clusters. The model generated



three clusters respectively for fake news, replica products and pharmaceuticals and pleasure medicines. All of this allowed to identify and understand how does Phishing emails are structured and composed.

## 13 Appendices

### A. Repository

The elaborated Model can be reviewed at the public repository published on *Github*, which is available at the following URL: **Github Model Repository**.

Inside the repository the divisions are as follows:

- **Readme file:** Description of the Github Repository
- **Preliminary Model Folder:** Folder Containing the five preliminary models elaborated with K-Means, DBSCAN, HDBSCAN, Spectral Clustering and Gaussian Mixture Models.
- **Final Model Folder:** Folder Containing the final model
- **GUI Folder:** Folder Containing the backend and the frontend for the presentation layer of the model.

### B. Model Input Data

The preliminary model data can be found on the following site: **Kaggle Phishing Dataset**

### C. Model Input Data

The model data can be found on the following site: **Kaggle Phishing Dataset**

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